PHONG NGUYEN

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Education

University of South Florida **May 2026**
Bachelor of Science in Computer Science *Tampa, FL*

Relevant Coursework

- Data Structures
- Algorithms Analysis
- Database Design
- Computer Architecture
- Artificial Intelligence
- Operating System
- Object-Oriented Programming
- Software Engineering
- Secure Coding

Projects

AI Short Video Generator | *Next.js, TypeScript, Supabase, Inngest, Antigravity* **Feb 2026 - Mar 2026**

- Engineered a fully automated video generation pipeline orchestrating Google GenAI for scriptwriting, Replicate for media synthesis, and Deepgram for high-fidelity text-to-speech, rendered programmatically via serverless Remotion Lambdas.
- Accelerated the full-stack development lifecycle utilizing AI-assisted rapid prototyping (Antigravity) to architect a scalable Next.js application, leveraging Inngest for reliable, asynchronous background orchestration of resource-intensive rendering tasks
- Integrated Clerk and Supabase for end-to-end SaaS functionality, establishing secure user authentication, protected routing, and real-time data persistence to support a multi-tiered, credit-based user generation model.

AI Voice Agent Interview | *Next.js, TypeScript, Prisma, PostgreSQL, Clerk* **Nov 2025 – Dec 2025**

- Engineered a full-stack, real-time voice agent platform using WebSockets and Node.js to facilitate simulated technical interviews, achieving sub-100ms average latency for bi-directional voice and text communication.
- Integrated the Gemini API for sophisticated Natural Language Processing (NLP) to dynamically evaluate user response accuracy and complexity, which allowed the agent to adjust the interview path in real time.
- Designed a modular feedback system utilizing Firebase for secure data persistence, generating immediate, personalized critiques (pace, technical depth) that significantly enhanced the learning experience for users.

AI SaaS Website Builder | *Next.js, React, TypeScript, Prisma, PostgreSQL* **Oct 2025 – Nov 2025**

- Engineered an AI agent using Inngest and OpenAI to generate and build full Next.js applications from text prompts within secure, containerized E2B sandboxes.
- Implemented a full-stack, type-safe architecture using tRPC for API communication and Prisma ORM for data persistence with PostgreSQL
- Integrated Clerk to build end-to-end SaaS functionality, including secure user authentication, protected routes, and a multi-tiered subscription model with credit-based rate limiting

Stock Chatbot | *Python, Flask, spaCy, Keras/TensorFlow, scikit-learn* **August 2025**

- Engineered a full-stack AI chatbot using Python and Flask, leveraging spaCy for natural language understanding (NLU) within a RESTful API to process user queries and serve livestock predictions
- Implemented an LSTM time series forecasting model with an automated retraining pipeline, ensuring sustained prediction accuracy on real-time data.

Finance Portfolio Dashboard | *Python, Flask, SQLAlchemy, yfinance, Chart.js* **July 2025**

- Built a full-stack dashboard (Flask, SQLAlchemy) using the yfinance API and Pandas to manage transactions and analyze portfolio performance.
- Engineered a responsive UI with Chart.js for live, asynchronous visualization of asset allocation and value.

Movie Review Sentiment Analysis | *Python, Scikit-learn, NLTK.* **April 2025**

- Developed a machine learning model to perform sentiment analysis on movie reviews, achieving 100% accuracy, precision, and recall on a balanced test set of over 2,500 reviews..
- Engineered a text preprocessing pipeline using NLTK for lemmatization, stop-word removal, and cleaning of raw text data.
- Implemented n-gram (unigram and bigram) feature extraction and trained a Logistic Regression classifier, validating the model's perfect performance with 5-fold cross-validation.

Technical Skills

Languages: Python, C/C++, JavaScript, TypeScript, SQL, HTML/CSS
Frameworks & Libraries: React, Next.js, Node.js, Django, Flask, FastAPI, TailwindCSS, SpaCy, Keras/TensorFlow
Databases: PostgreSQL, MySQL, MongoDB, SQLite, Prisma, SQLAlchemy
DevOps & Tools: Git, Github, Docker, Vercel, Clerk, Postman, Jira, OpenAI API, Codex, Claude Code